

Towards Practical LLM Unlearning: Efficient, Modular, and Retain-Free

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Abstract

Large Language Models (LLMs), trained on vast web corpora and now widely integrated into web services like search engines and chatbots, have raised growing concerns about their privacy and security. This has spurred the development of Machine Unlearning techniques, which aim to effectively remove the influence of specific data (such as private user information, copyrighted web content, or harmful knowledge) from trained models while preserving general performance. However, existing LLM unlearning approaches face significant challenges, including extensive parameter modifications, a heavy reliance on retain sets to preserve utility, and limited scalability for handling sequential unlearning requests, impeding their real-world deployment in dynamic web environments. In this work, we propose Semantic Redirection for Unlearning (SRU), a light-weight framework that fine-tunes only the output word embedding layer. This targeted adjustment reshapes the model’s semantic-to-lexical mapping to block undesired concepts without disturbing deeper representations, thereby preserving the model’s general performance and eliminating dependency on retain sets. Furthermore, SRU’s modular design enables independent, sequential unlearning tasks—a vital feature for live web services handling continuous data removal requests. Experiments on the MUSE benchmark demonstrate that SRU achieves state-of-the-art efficiency, reducing

computational cost by approximately 98%, while maintaining competitive unlearning performance, making it a practical and efficient solution for building compliant and trustworthy LLM-powered web applications.

CCS Concepts

• **Security and privacy** → **Privacy protections; Web application security.**

Keywords

Machine Unlearning; Large Language Model; Privacy protection

ACM Reference Format:

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1 Introduction

Large Language Models (LLMs), trained on massive and diverse text corpora scraped from the public web, are now integral to a new generation of web applications, from intelligent search assistants to automated content creation tools. This deep entanglement with the Web ecosystem creates a critical privacy risk: sensitive information originating from the web becomes embedded within these models, which are then deployed as services that can expose this data back to web users. Consequently, they may inadvertently memorize and reveal personally identifiable information (PII) harvested from social media profiles or forums [10, 13, 27], or reproduce copyrighted text from online articles when subjected to probing or jailbreak

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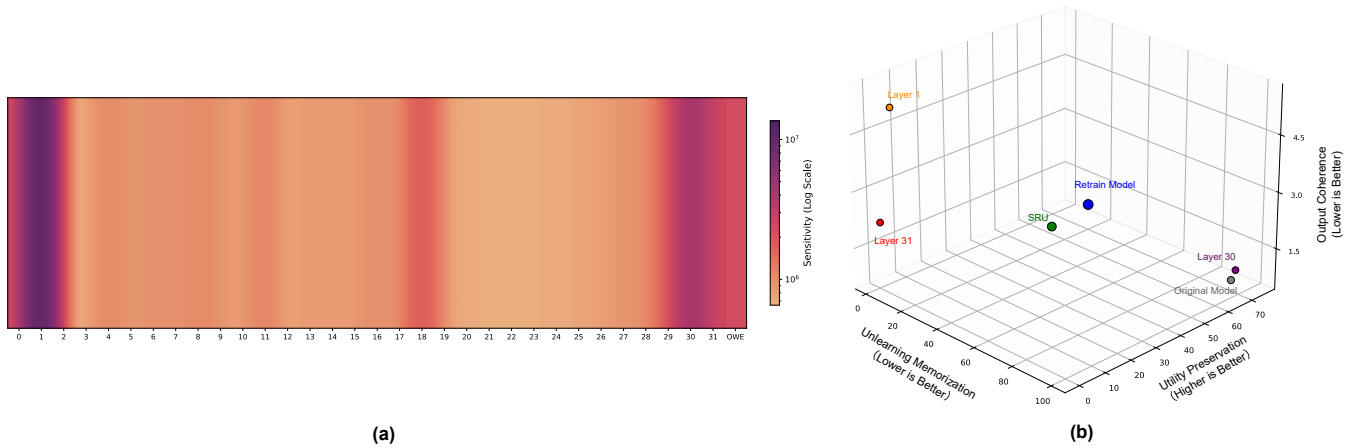


Figure 1: (a) Characterizing Layer Responsiveness via Gradient Changes with Forget Data. Darker regions indicate higher sensitivity. (b) Evaluating layer adjustment strategies on Unlearning Memorization (x-axis), Utility Preservation (y-axis), and Output Coherence (z-axis). Adjustments were made individually to highly sensitive layers identified in (a). Modifying Layer 1 or Layer 31 severely degrades utility and coherence. Adjusting Layer 30 shows minimal unlearning effectiveness (close to Original Model), making these specific layer adjustments less viable compared to targeting the output word embedding layer.

attacks [6, 16, 46, 51]. Moreover, biases and harmful content prevalent on the web can be amplified, leading to inappropriate outputs [9]. Regulations like the EU’s GDPR, which introduces the "Right to be Forgotten", create a legal mandate for operators of these web platforms to remove user data upon request, underscoring the urgent need for techniques to safely remove sensitive information from the LLMs that power these platforms [45, 50].

Machine unlearning aims to selectively remove specific data and its influence from a trained model, while preserving the model’s general performance on remaining data (i.e., model utility) [14, 33, 39, 43, 49]. While exact unlearning, which entails retraining the model from scratch on a sanitized dataset, provides the strongest theoretical guarantees of data removal [5], it is computationally prohibitive for LLMs [34]. As a result, research has increasingly focused on approximate unlearning approaches, which aim to efficiently modify a model’s parameters (e.g., through fine-tuning) to closely approximate the behavior of a model retrained from scratch. However, prevailing approximate methods, such as Gradient Ascent (GA) [21] and Negative Preference Optimization (NPO) [52], often suffer from critical limitations. For instance, these approaches frequently necessitate large-scale parameter modifications, leading to inefficiency. Moreover, studies have shown that such methods can face a significant reduction in model utility when handling sequential unlearning requests. Critically, they commonly rely on retain sets to prevent catastrophic forgetting and preserve model utility [24, 36]. The practical acquisition and maintenance of suitable, often large, retain sets pose significant challenges, slowing down the unlearning process and increasing deployment complexity, thereby hindering their real-world applicability.

The challenge, therefore, lies in identifying an intervention point that is both effective for unlearning and minimally disruptive to overall model utility. Our analysis of layer responsiveness offers insights into this problem. As depicted in Figure 1(a), gradient changes in response to unlearning signals are primarily concentrated in the

initial layers (early Transformer layers) and the final layers (later Transformer layers and the output word embedding layer). While some core Transformer layers might exhibit high gradient responsiveness, interventions there carry a substantial risk of disrupting the intricate web of learned features crucial for general language understanding, as knowledge in these layers is often highly entangled [4, 18, 42]. In contrast, the output word embedding layer, acting as the final vocabulary mapping, offers a more specific and disentangled point of intervention (Figure 1(b)) [30, 32]. This layer is critical in the language model generation pipeline, responsible for translating the model’s final internal semantic representations (i.e., context vectors) into probability distributions over the vocabulary [7]. Modifying it primarily alters how pre-existing internal representations are expressed into concrete lexical outputs, rather than fundamentally changing those core representations themselves [48]. Moreover, the inherent preservation of deeper knowledge, along with the targeted and minimal nature of parametric modifications, forms the foundation for maintaining model utility without relying on retain set.

Building on these insights, we introduce **Semantic Redirection for Unlearning (SRU)**. SRU fine-tunes only the output word embedding layer using a contrastive data-augmentation mechanism: perturbed variants of the forget data act as positive examples, while the original data serve as negatives. This dual-example approach steers the model’s semantic-to-lexical mapping away from undesired concepts, effectively blocking the generation of specific words or ideas tied to the forgotten data. By restricting updates to this single interface layer, SRU leaves deeper parameters untouched—maintaining general capabilities and avoiding the high computational cost of broad parameter updates. The result is a precise, efficient, and low-impact unlearning method that removes sensitive knowledge while preserving overall model utility. A further advantage of SRU is its modular design: the updated output embedding layer functions as an independent, plug-and-play component. This plug-and-play

nature inherently supports multiple, sequential unlearning tasks on the same base model, enabling efficient handling of successive forget requests.

We conducted comprehensive experimental validation on two prominent benchmarks, the Machine Unlearning Six-Way Evaluation (MUSE) [41] and the Task of Fictitious Unlearning (TOFU) [34], to thoroughly assess the performance of our method. The results robustly demonstrate SRU’s effectiveness: without requiring access to retain data, SRU achieves competitive unlearning performance while preserving model utility and yielding substantial improvements in computational efficiency. Notably, SRU yields a substantial 98% improvement in computational efficiency compared to representative baseline methods. By achieving a compelling combination of competitive unlearning performance and state-of-the-art (SOTA) efficiency on the benchmarks while successfully resolving common drawbacks of previous approaches—namely dependence on retain sets and inability to handle sequential requests—SRU significantly enhances the practical applicability of LLM unlearning techniques.

2 Related Work

2.1 Machine Unlearning

Machine Unlearning aims to remove specific data and its influence from trained models. Research in this field initially focused on tasks such as image classification and continues to explore various strategies [15], including exact unlearning (e.g., aggregating models trained on data partitions [3]) and approximate unlearning (e.g., estimating data influence based on Hessian matrix analysis [2, 17, 29, 44]). However, for LLMs, the direct application of these foundational unlearning methods that might be effective in other domains often faces severe efficiency challenges due to their substantial computational and storage overheads [28].

Consequently, research on unlearning for LLMs has predominantly shifted towards more adaptive approximate unlearning, typically implemented through fine-tuning pre-trained models. Current mainstream methods include: GA and its variants (e.g., combined with retention loss or KL divergence constraints) for adversarial optimization on target data to weaken model memory while maintaining general performance [21, 31, 47]; employing NPO to adjust model preferences [52]; or intervening in output logits via auxiliary models [23]. However, a common drawback of these approaches is their inherent inefficiency for rapid unlearning, as they necessitate large-scale parameter modifications, leading to substantial computational overhead.

2.2 Word Embedding

Word Embedding is a core component in modern Natural Language Processing (NLP) and language models [25]. It maps discrete vocabulary symbols into low-dimensional, dense vector spaces, thereby capturing semantic and syntactic relationships between words. The study of the structure and properties of word embedding spaces has been crucial for understanding the internal mechanisms of language models. Early research, such as methods based on Word2Vec, revealed potential linear relationships between word embedding vectors (e.g., the famous ‘king - man + woman = queen’ analogy) [1, 8]. Subsequent studies explored the properties of word embedding spaces from more complex geometric perspectives, such as

rotational invariance and anisotropy [12, 37]. Concurrently, substantial work has analyzed how word embedding vectors represent word meanings from the perspective of lexical semantics [22, 35, 40]. However, most research on word embeddings has primarily focused on their representational capacity and static properties. Recently, research pays more attention to the role of word embeddings in the generation process of language models. For example, Han et al. [19] explore the connection between word embeddings and model generation behavior.

3 Method

3.1 Problem Definition and Objective

In the setting of Machine Unlearning, let D represent the original complete training dataset. This dataset can be divided into a data subset to be forgotten, D_f (Forget Set), and a data subset to be retained, $D_r = D \setminus D_f$ (Retain Set). The original model trained on the complete dataset D , denoted as f_{target} (Target Model), is the initial state for the unlearning task. Ideally, we aim to obtain a model trained solely on the retain set D_r , denoted as the ideal retrained model f_{retrain} (Retrained Model). The goal of machine unlearning is to design an unlearning algorithm U that adjusts the original model f_{target} into a new model $f_{\text{unlearn}} = U(f_{\text{target}})$, such that f_{unlearn} behaves as closely as possible to f_{retrain} . Given that achieving such Exact Unlearning ($f_{\text{unlearn}} = f_{\text{retrain}}$) is typically computationally prohibitive, practical research primarily pursues Approximate Unlearning, which aims to ensure that f_{unlearn} closely approximates f_{retrain} in terms of model behavior (e.g., prediction outputs, internal representation distributions), i.e., $f_{\text{unlearn}} \approx f_{\text{retrain}}$.

3.2 Semantic Redirection for Unlearning

LLMs typically generate text sequences in an auto-regressive manner, predicting the probability distribution of the next word o_t given the previously generated sequence $(o_1, \dots, o_{t-1})$, i.e., $P(o_t | o_1, \dots, o_{t-1})$. The standard implementation process is as follows: First, the model computes a corresponding context representation vector $\mathbf{c}(o_1, \dots, o_{t-1})$ based on the preceding information $(o_1, \dots, o_{t-1})$. Subsequently, this context vector is operated upon with the model’s output word embedding matrix $\mathbf{E} \in \mathbb{R}^{d \times |O|}$ (where $\mathbf{E} = [\mathbf{e}_{o_1}, \dots, \mathbf{e}_{o_{|O|}}]^\top$, d represents the embedding dimension, and $|O|$ denotes the vocabulary size) to obtain the logits for each word in the vocabulary: $\mathbf{l} = \mathbf{c}(o_1, \dots, o_{t-1})^\top \mathbf{E}$. Finally, these logits are converted into a probability distribution using the Softmax function [53]. The output word embedding matrix $\mathbf{E}$ is learned during the model’s pre-training phase and is responsible for mapping the model’s internal hidden states (i.e., the context vector $\mathbf{c}$) to the vocabulary space, thereby directly determining the final word selection probabilities. Layer responsiveness analyses indicate that while initial and final layers show high gradient sensitivity to forget data, the output word embedding layer provides a more specific and disentangled point of intervention compared to the substantial risk of disrupting highly entangled core Transformer layers. Modifying this layer thus offers a precise and minimally disruptive approach to control generation, as it primarily reshapes how pre-existing knowledge is expressed rather than altering the core representations themselves. This aligns with existing research (e.g., Han et al. [19]) demonstrating that applying linear transformations to the output word

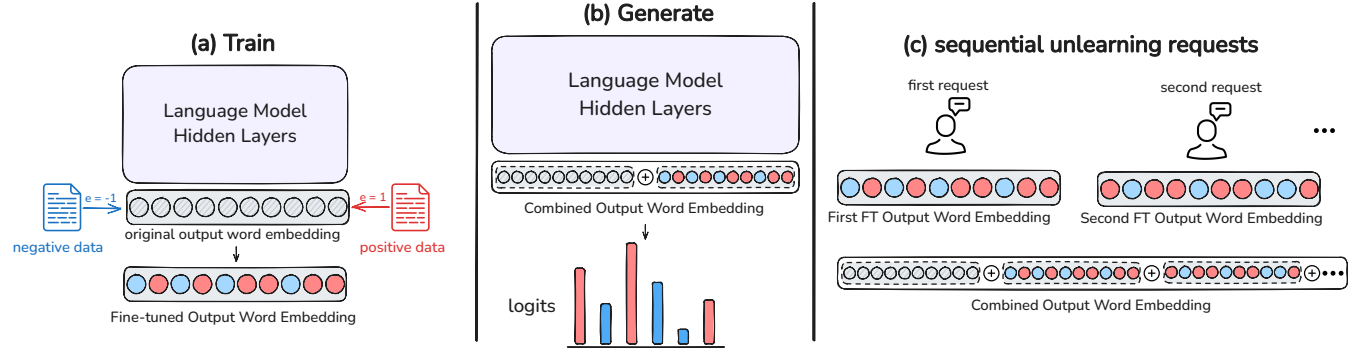


Figure 2: Overview of the SRU framework. (a) Training Phase: A correction embedding matrix acting on the output word embedding layer is trained using positive samples (perturbed forget data) and negative samples (original forget data). (b) Generation Phase: The learned unlearning embedding is linearly combined with the original output word embedding layer to guide the model generation process, avoiding outputs related to the forgotten content. (c) Sequential Unlearning Handling: For each independent unlearning request, a corresponding correction embedding matrix is trained. During inference, all relevant correction embedding matrices are aggregated with the original embedding matrix to respond to sequential unlearning demands.

embedding matrix can effectively modulate the model's output distribution. Building on these observations, we propose a novel machine unlearning method—SRU. The core idea of this method is to learn a dedicated correction embedding matrix E' and combine it with the original output word embedding matrix E to adjust the model's output probability distribution, thereby suppressing the model's tendency to generate content related to the forget set, and thus achieving approximate unlearning.

3.3 Training Process of SRU

The training process of SRU draws inspiration from the ideas of Contrastive Learning [26] and Preference Optimization [38], but employs more direct supervisory signals in its specific implementation. We construct two types of data samples: "Positive Examples," which represent output content the model should prefer to generate, and "Negative Examples," which represent output content the model should avoid generating and needs to forget (Examples are provided in Appendix). As shown in Figure 2(a), the training of SRU focuses on learning a correction embedding matrix E' that acts on the output word embedding layer, while the main body of the model (including the original embedding matrix E and parameters of other parts) remains frozen during this process. This strategy significantly reduces the number of parameters to be trained, thereby substantially improving the computational efficiency of the unlearning process.

During the training phase, the objective is to optimize the correction embedding matrix E' so that it learns an effective adjustment signal. When this signal (via E') is combined with the original embedding matrix E , it can guide the model to produce behavior consistent with the unlearning objectives. We use the standard cross-entropy loss function for optimization. Specifically, for a given context x and the true next word y , the logits computed by the original model are $I(x) = c(x)^T E$, and the correction logits generated by the correction embedding matrix are $I'(x) = c(x)^T E'$. The training objective is to optimize the parameters of E' .

We define different training objectives for positive and negative samples, distinguishing them (implicitly) by their source:

For positive samples $(x_{\text{pos}}, y_{\text{pos}}) \in D_f'$ (composed of perturbed forget data): the goal is to guide the model towards generating distributions represented by the positive text. Therefore, we minimize the standard cross-entropy loss for predicting the target word y_{pos} on the combined logits $I(x_{\text{pos}}) + \lambda I'(x_{\text{pos}})$. Its loss function is defined as:

$$\mathcal{L}_{\text{pos}} = \text{CE}(\text{softmax}(I(x_{\text{pos}}) + \lambda I'(x_{\text{pos}})), y_{\text{pos}}) \quad (1)$$

where CE denotes the cross-entropy loss, and λ is a hyperparameter set during training to control the strength of the correction signal.

For negative samples $(x_{\text{neg}}, y_{\text{neg}}) \in D_f$ (representing content to be forgotten): the goal is to suppress the model from generating the specific output y_{neg} associated with the forget data. We achieve this by maximizing the cross-entropy loss for predicting the target word y_{neg} on the combined logits $I(x_{\text{neg}}) + \lambda I'(x_{\text{neg}})$ (equivalent to minimizing its negative value). This drives E' to learn a correction signal that counteracts or weakens the tendency of the original logits $I(x_{\text{neg}})$ to predict y_{neg} . Its loss function is defined as:

$$\mathcal{L}_{\text{neg}} = -\text{CE}(\text{softmax}(I(x_{\text{neg}}) + \lambda I'(x_{\text{neg}})), y_{\text{neg}}) \quad (2)$$

The total loss function $\mathcal{L}(E')$ is a weighted sum of these two parts, minimized using optimization algorithms such as gradient descent, updating only the parameters of E' :

$$\begin{aligned} \mathcal{L}(E') = & \sum_{(x_{\text{pos}}, y_{\text{pos}}) \in D_f'} \mathcal{L}_{\text{pos}}(x_{\text{pos}}, y_{\text{pos}}) \\ & + \beta \sum_{(x_{\text{neg}}, y_{\text{neg}}) \in D_f} \mathcal{L}_{\text{neg}}(x_{\text{neg}}, y_{\text{neg}}) \end{aligned} \quad (3)$$

where β is a hyperparameter that balances the importance of the desired data (positive examples) against the suppression strength

for the forget data (negative examples). By optimizing this total loss, E' is trained to encode the necessary adjustment information, such that the final combined embedding $E_{\text{unlearn}} = E + \lambda E'$ can effectively suppress outputs related to the forget data while preserving the model's capabilities on general data as much as possible.

3.4 Deployment and Extensibility of SRU

The correction output word embedding matrix E' obtained through SRU training exhibits significant modularity. As show in Figure 2(b), during the generation stage, it can be used as a "plug-and-play" component in combination with the original embedding matrix E of the target model f_{target} . The effective word embedding matrix E_{unlearn} used for generating logits is formed through the following additive combination:

$$E_{\text{unlearn}} = E + \lambda_{\text{inf}} E' \quad (4)$$

Here, λ_{inf} is a hyperparameter used during the inference phase, controlling the application strength of the unlearning correction E' . The value of λ_{inf} can be adjusted based on the unlearning performance on a validation set. When $\lambda_{\text{inf}} = 0$, the model behaves identically to the original model; a non-zero value for λ_{inf} introduces the unlearning adjustment. (Note: The λ used during the training phase and the λ_{inf} used during the inference phase can be the same or different, with the latter offering greater flexibility for adjustment.)

Furthermore, SRU's modular design naturally supports handling sequential unlearning requests, making it particularly suitable for scenarios requiring responses to multiple independent unlearning tasks in close temporal proximity. When the model receives multiple unlearning requests corresponding to different forget datasets $D_{f_1}, D_{f_2}, \dots, D_{f_k}$, a respective correction embedding matrix $E'_1, E'_2, \dots, E'_k$ can be independently trained for each request. As show in Figure 2(c), during deployment, the effects of multiple unlearning tasks can be superimposed by performing a weighted sum of these independent correction matrices with the original embedding matrix E :

$$E_{\text{unlearn}} = E + \sum_i \lambda_i E'_i \quad (5)$$

where λ_i is the strength control parameter associated with the i -th unlearning task. This linear combination approach provides an efficient, flexible, and scalable solution for responding to dynamic and dense unlearning demands.

4 Experiment

4.1 Experimental Setup

Evaluation Benchmark. To comprehensively and systematically evaluate the effectiveness of our proposed SRU unlearning method, we adopt the MUSE benchmark [41]. Recognized for its comprehensive evaluation capabilities, MUSE serves as a common yardstick in the LLM unlearning community, facilitating robust comparisons across various methods. It provides a multi-dimensional perspective for evaluating the unlearning of LLMs, measuring performance based on several key attributes crucial for real-world deployment.

These attributes consider the concerns of both Data Owners and Model Deployers.

Metrics concerning Data Owners include: Verbatim Memorization Removal, which measures the removal of exact reproductions of content, typically assessed using the ROUGE-L score on the Forget Set; Knowledge Memorization Removal, which measures the removal of related knowledge, quantified by calculating the ROUGE-L score on derived question-answering pairs; and Privacy Protection, evaluating the risk of privacy leakage, often measured using Membership Inference Attacks (MIA).

Metrics concerning Model Deployers include: Utility Preservation, which measures the model's general capability, typically reflected by evaluating knowledge memorization on the Retain Set; and Sustainability, assessing the model's performance when handling sequential unlearning requests, evaluated by tracking the trends of the aforementioned core metrics. A notable feature of the MUSE benchmark is its use of challenging real-world datasets (such as news, books) rather than synthetic data, providing a more rigorous and realistic testbed for unlearning method performance.

Model and Baselines. To ensure fair comparison, all experiments in this study use the pre-trained language model provided by the MUSE benchmark as the initial model for the unlearning task. For baseline methods, we selected representative methods already evaluated in the MUSE benchmark for comparison, including GA [47], NPO [52], Task Vector (TV) [20], and Who's Harry Potter (WHP) [11]. The evaluation process strictly adheres to the standardized evaluation scripts and datasets provided by MUSE (including the Target Model, Retrain Model, and results data for each baseline method) to guarantee the credibility and reproducibility of the evaluation results. Additionally, some baseline methods might incorporate extra utility preservation mechanisms, such as Gradient Descent on the Retain set to reduce performance loss (GDR) and using KL divergence constraints to keep the unlearned model's distribution close to the original model (KLR); we explicitly indicate these variants when reporting results. To further validate the robustness of our method, we also conduct evaluations on the TOFU [34] benchmark, with detailed results presented in Section 4.3.

4.2 Evaluation Results on MUSE

Table 1 clearly presents a detailed performance comparison of the SRU method against various baseline models on the MUSE benchmark. To ensure the practical relevance of the evaluation, baseline methods that caused model utility degradation exceeding 40% on the retain set compared to the Retrain Model are distinguished by gray text in Table 1. Due to their limited usability in real-world scenarios, these highlighted methods are generally de-emphasized in our subsequent analysis of practical performance. Furthermore, it is important to note that the 'No Privacy Leak' metric, though a component of the MUSE benchmark, is not a primary focus of our discussion in this section. This is because effectively mitigating privacy leakage remains a common and significant challenge for nearly all current unlearning methods, including those assessed in this work.

On the knowledge memory removal metric, SRU demonstrated excellent performance, achieving optimal results on both the NEWS

Table 1: Performance comparison on the MUSE dataset. Methods shown in gray text indicate baseline approaches where model utility degrades by more than 40% compared to the Retrain Model.

	No Verbatim Mem. <i>VerbMem on D_{forget} ($\downarrow$)</i>		No Knowledge Mem. <i>KnowMem on D_{forget} ($\downarrow$)</i>		No Privacy Leak. <i>PrivLeak ($\in [-5\%, 5\%]$)</i>		Utility Preserv. <i>KnowMem on D_{retain} ($\uparrow$)</i>	
NEWS								
Target f_{target}	58.4		63.9		-99.8		55.2	
Retrain f_{retrain}	20.8		33.1		0.0		55.0	
GA	0.0	$\downarrow 100\%$	0.0	$\downarrow 100\%$	5.2	over-unlearn	0.0	$\downarrow 100\%$
GA _{GDR}	4.9	$\downarrow 76.5\%$	31.0	$\downarrow 6.3\%$	108.1	over-unlearn	27.3	$\downarrow 50.3\%$
GA _{KLR}	27.4	$\uparrow 31.4\%$	50.2	$\uparrow 51.5\%$	-96.1	under-unlearn	44.8	$\downarrow 18.5\%$
NPO	0.0	$\downarrow 100\%$	0.0	$\downarrow 100\%$	24.4	over-unlearn	0.0	$\downarrow 100.0\%$
NPO _{GDR}	1.2	$\downarrow 94.4\%$	54.6	$\uparrow 64.8\%$	105.8	over-unlearn	40.5	$\downarrow 26.3\%$
NPO _{KLR}	26.9	$\uparrow 29.0\%$	49.0	$\uparrow 48.1\%$	-95.8	under-unlearn	45.4	$\downarrow 17.4\%$
TV	57.2	$\uparrow 174.7\%$	66.2	$\uparrow 100.0\%$	-99.8	under-unlearn	55.8	$\uparrow 1.5\%$
WHP	19.7	$\downarrow 5.6\%$	21.2	$\downarrow 35.9\%$	109.6	under-unlearn	28.3	$\downarrow 48.5\%$
SRU	26.2	$\uparrow 26.0\%$	32	$\uparrow 3.3\%$	-99.7	under-unlearn	39.2	$\downarrow 28.7\%$
BOOKS								
Target f_{target}	99.8		59.4		-57.5		66.9	
Retrain f_{retrain}	14.3		28.9		0.0		74.5	
GA	0.0	$\downarrow 100\%$	0.0	$\downarrow 100\%$	-25.0	under-unlearn	0.0	$\downarrow 100\%$
GA _{GDR}	0.0	$\downarrow 100\%$	0.0	$\downarrow 100\%$	-26.5	under-unlearn	10.7	$\downarrow 85.6\%$
GA _{KLR}	16.0	$\uparrow 11.4\%$	21.9	$\downarrow 24.4\%$	-40.2	under-unlearn	37.2	$\downarrow 50.0\%$
NPO	0.0	$\downarrow 100\%$	0.0	$\downarrow 100\%$	-24.3	under-unlearn	0.0	$\downarrow 100\%$
NPO _{GDR}	0.0	$\downarrow 100\%$	0.0	$\downarrow 100\%$	-30.8	under-unlearn	22.8	$\downarrow 69.4\%$
NPO _{KLR}	17.0	$\uparrow 18.2\%$	25.0	$\downarrow 13.4\%$	-43.5	under-unlearn	44.6	$\downarrow 40.1\%$
TV	99.7	$\uparrow 595.0\%$	52.4	$\uparrow 81.2\%$	-57.5	under-unlearn	64.7	$\downarrow 13.1\%$
WHP	18.0	$\uparrow 25.2\%$	55.7	$\uparrow 92.9\%$	56.5	over-unlearn	63.6	$\downarrow 14.6\%$
SRU	20.7	$\uparrow 44.8\%$	21.5	$\downarrow 25.6\%$	-50.7	under-unlearn	54.2	$\downarrow 27.2\%$

and BOOKS subsets of the dataset. Specifically, on the NEWS subset, its score was only 3% higher than the theoretically optimal Retrain Model; whereas on the BOOKS subset, its score was even lower than the Retrain Model (note: a lower score on this metric indicates a better knowledge removal effect). These results fully demonstrate the significant effectiveness of SRU in eliminating the model’s memory of specific knowledge points.

For the verbatim memory removal metric, SRU achieved sub-optimal results. This phenomenon indicates that for SRU, which only adjusts parameters in the output embedding layer, completely removing the model’s precise verbatim memory of long text sequences remains somewhat challenging. Although its performance on this metric was not optimal, SRU’s score was still significantly lower than the original Target Model, indicating that it can still achieve an effective forgetting effect.

In terms of model utility preservation, SRU exhibited competitive performance without leveraging a retain set for additional optimization. Compared to baseline methods that involve more extensive adjustments to the model’s internal parameters, SRU performed better in preserving model utility. Although its performance was slightly inferior to methods that use retain data for optimization, this instead highlights SRU’s practical application potential and

advantages in scenarios where resources are limited or access to a retain set is unavailable.

In a comprehensive analysis, SRU achieved a more desirable balance between key forgetting performance metrics (especially knowledge memory removal) and model utility preservation, effectively avoiding significant performance degradation in any single dimension. Particularly important is that the aforementioned competitiveness demonstrated by SRU was achieved while significantly improving computational efficiency and substantially reducing resource consumption (for details, see Section 5.1). This characteristic is of crucial significance for enhancing the feasibility of machine unlearning techniques in practical deployment and application.

4.3 Evaluation Results on TOFU

To further assess the robustness and generalizability of the proposed SRU method, we conducted additional experiments on the TOFU benchmark. Unlike the MUSE benchmark, which uses real-world data, TOFU is specifically designed to evaluate a model’s ability to unlearn fictional author biographies after being fine-tuned on a question-answering dataset. The benchmark sets three ascending forget ratios—1%, 5%, and 10%—which correspond to unlearning tasks of varying difficulty. Regarding evaluation metrics, the experiments employ ROUGE-L to quantify Forgetting Performance, while

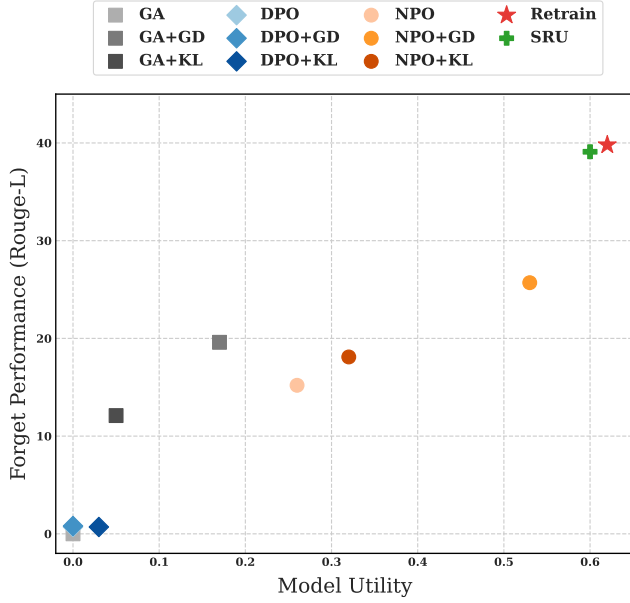


Figure 3: Performance on TOFU-10%. The top-right region indicates a better trade-off between forgetting and utility, with the Retrain model as the target.

Model Utility is assessed by combining the model’s performance on tasks related to fictitious authors, real-world authors, and general world knowledge. To ensure a fair comparison, all experiments strictly adhere to the pre-trained models and evaluation scripts officially provided by TOFU.

The results (depicted in Figure 3) indicate that SRU also demonstrates excellent performance on the TOFU benchmark. Specifically, under the 10% forget ratio setting, SRU achieves an optimal balance between the key metrics of Forgetting Performance and Model Utility. The complete results for all tested forget ratios (1%, 5%, and 10%) are detailed in the Appendix. Given that the TOFU benchmark differs significantly from MUSE in its data domain and task paradigm, SRU’s success on this benchmark further validates that its effectiveness is not limited to specific configurations. This result underscores the robustness and practical application potential of our unlearning method.

5 Additional Analyses

5.1 Efficiency analysis

To evaluate the training efficiency of SRU, we compared it with baseline methods on MUSE. The key evaluation metric was the wall-clock time required for a single unlearning gradient update step and the total unlearning time, detailed data are shown in Table 2. To ensure a fair comparison, all experiments were conducted on an NVIDIA A40 GPU hardware platform. Notably, during the time measurement process, the baseline method utilized 8 NVIDIA A40 GPUs for parallel computation, whereas the SRU method only required a single NVIDIA A40 GPU to complete the same task. Apart from the difference in the number of GPUs, the batch size and input sequence length in the experiments were kept consistent

Table 2: Wall-clock time and total GPU hours required for each unlearning method.

Method	GPU Use	Time (Seconds/Step)	Total Time (GPU Hours)
Retrain	8	-	184320
GA	8	4.14	0.56
GA _{GDR}	8	6.05	0.82
GA _{KLR}	8	7.58	1.03
NPO	8	5.68	0.77
NPO _{GDR}	8	7.59	1.03
NPO _{KLR}	8	9.11	1.24
TV	8	4.14	1.12
WHP	8	4.14	1.12
SRU	1	1.36	0.08

with the baseline method, set to 32 and 2048 tokens, respectively. Therefore, the total number of tokens processed in each gradient update step was $32 \times 2048 = 65536$ tokens.

In terms of hardware resource utilization, SRU’s use of only a single GPU significantly reduces the deployment threshold and operational costs of the method. Regarding the time per step, the baseline method, utilizing 8 GPUs in parallel, achieved an optimal time per step of 4.14 seconds. Despite this, SRU, operating on only a single GPU, was still approximately 3 times faster per step. More importantly, when considering the Total Compute Time, the total computation required for SRU to complete the entire unlearning process was only 0.08 GPU hours, significantly lower than the range of 0.56 to 1.24 GPU hours for the baseline methods.

In summary, the experimental results strongly demonstrate that the SRU method possesses significant advantages in resource utilization efficiency and computational speed. This characteristic makes SRU exhibit excellent applicability and application potential in practical scenarios with limited computational resources or those requiring rapid model iteration.

5.2 Analysis of Sequential Unlearning Requests

To evaluate SRU’s capability in handling sequential unlearning requests, this study uniformly divided the MUSE unlearning set into 1-4 subsets to simulate multiple independent unlearning requests arriving sequentially in real-world scenarios. For each simulated unlearning request (i.e., each unlearning subset), we independently trained a corresponding unlearning embedding. During the model inference phase, these independently trained unlearning embeddings were linearly combined to collectively affect the model’s output. Table 3 presents the performance of SRU when handling 1-4 sequential unlearning requests, respectively. The experimental results indicate that when faced with such sequential unlearning tasks, SRU’s key performance metrics remain stable, without exhibiting significant degradation.

5.3 Data Usage Ablation

Many machine unlearning methods rely on an independent retain set to maintain the model’s general performance. The design of SRU

Table 3: Performance evaluation of SRU when handling sequential unlearning requests.

	Seq. Num	Verb. Mem. $D_{\text{forget}} (\downarrow)$		Know. Mem. $D_{\text{forget}} (\downarrow)$		Utility $D_{\text{retain}} (\uparrow)$	
NEWS							
Target	-	58.4		63.9		55.2	
Retrain	-	20.8		33.1		55.0	
SRU	1	26.2	$\uparrow 26.0\%$	32	$\uparrow 3.3\%$	39.2	$\downarrow 28.7\%$
	2	26.4	$\uparrow 26.9\%$	35.9	$\uparrow 8.4\%$	40.3	$\downarrow 26.7\%$
	3	25.3	$\uparrow 21.8\%$	34.1	$\uparrow 3.0\%$	41.2	$\downarrow 25.1\%$
	4	25.0	$\uparrow 20.2\%$	34.0	$\uparrow 2.9\%$	39.3	$\downarrow 28.5\%$
BOOKS							
Target	-	99.8		59.4		66.9	
Retrain	-	14.3		28.9		74.5	
SRU	1	20.7	$\uparrow 44.8\%$	21.5	$\downarrow 25.6\%$	54.2	$\downarrow 27.2\%$
	2	26.8	$\uparrow 87.4\%$	33.6	$\uparrow 16.3\%$	53.0	$\downarrow 28.9\%$
	3	19.7	$\uparrow 37.4\%$	28.2	$\downarrow 2.5\%$	53.2	$\downarrow 28.6\%$
	4	35.7	$\uparrow 149.8\%$	25.6	$\downarrow 11.5\%$	53.7	$\downarrow 27.9\%$

Table 4: Impact of the data augmentation strategy on SRU.

	Data D_f'	Verb. Mem. $D_{forget} (\downarrow)$	Know. Mem. $D_{forget} (\downarrow)$	Utility $D_{retain} (\uparrow)$			
NEWS							
Target	-	58.4	63.9	55.2			
Retrain	-	20.8	33.1	55.0			
SRU	$\times$	27.2	$\uparrow 30.8\%$	37.7	$\uparrow 13.9\%$	29.6	$\downarrow 46.2\%$
SRU	$\checkmark$	26.2	$\uparrow 26.0\%$	32	$\uparrow 3.3\%$	39.2	$\downarrow 28.7\%$
BOOKS							
Target	-	99.8	59.4	66.9			
Retrain	-	14.3	28.9	74.5			
SRU	$\times$	91.4	$\uparrow 539.2\%$	25.9	$\downarrow 10.4\%$	48.8	$\downarrow 34.5\%$
SRU	$\checkmark$	20.7	$\uparrow 44.8\%$	21.5	$\downarrow 25.6\%$	54.2	$\downarrow 27.2\%$

avoids dependence on a retain set, aiming to enhance the method's practicality and ease of deployment. To achieve effective data contrast during the training phase, SRU defines the original data to be forgotten (forget data) as negative samples, and concurrently, versions of this forget data generated through semantic perturbation are used as positive samples. Specifically, these positive samples are generated by using an LLM to apply semantics-preserving perturbations to the original forget data. Table 4 presents the specific impact of this data augmentation strategy on SRU's performance. The experimental results show that after adopting this data contrast augmentation strategy, SRU achieved improvements in both the thoroughness of unlearning and the preservation of the model's general utility, thereby validating the effectiveness of this data contrast method.

5.4 Hyper-parameters analysis

For both the NEWS and BOOKS datasets, consistent training parameter configurations were adopted in this study. Specifically, the training hyperparameters were set as follows: a batch size of 1 and a learning rate of $1e-2$. During the training phase, for positive and negative samples, the corresponding adjustment factor λ was set to 1 and -1, respectively. It is worth noting that during the training process, the performance of the SRU model was observed to have a negative correlation with batch size; that is, a smaller batch size contributed to performance improvement. A potential explanation is that this phenomenon may stem from the intrinsic requirement of SRU for fine-grained adjustments to its internal mappings, as smaller batch sizes can support more precise gradient updates. Furthermore, experimental results indicate that the SRU model can achieve satisfactory performance levels with only a single training epoch. During the inference phase, the value of the adjustment factor λ can be adapted based on the specific application scenario. According to the experience from this study, setting λ to 1 or 2 typically yields optimal inference results.

6 Conclusion

This paper proposed an innovative LLM unlearning framework, SRU, which achieves efficient unlearning by precisely editing the output word embedding layer. SRU can effectively unlearn and maintain the model's general performance without relying on retain set, while significantly reducing computational overhead by approximately 98% and demanding lower hardware resources. Furthermore, its unique plug-and-play modular design supports sequential unlearning requests—a critical capability for live web platforms needing to address continuous data removal mandates like the "Right to be Forgotten." Extensive evaluations demonstrate the efficiency, flexibility, and practicality of SRU. By providing a scalable and lightweight solution, SRU offers a promising path for developing and maintaining safe, trustworthy, and compliant LLM-powered web applications, ultimately contributing to a more responsible Web ecosystem.

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A Limitations

Although the SRU framework proposed in this paper has made significant progress in improving the efficiency of unlearning training, its current implementation still has several limitations that warrant further discussion and refinement in future work.

First, the mechanism adopted in the SRU framework, where the original embedding matrix and the correction embedding matrix are stored separately and participate in computation, while key to achieving efficient unlearning, introduces additional computational overhead during the model inference phase, thereby impacting inference speed to a certain extent. However, this loss in inference efficiency can theoretically be optimized or eliminated by pre-merging the correction embedding matrix with the original embedding matrix into a single embedding matrix before inference. Second, although SRU is designed to effectively handle sequential unlearning requests, its performance and resource consumption may still face challenges when dealing with very large-scale or extremely long-sequence sequential unlearning tasks. Specifically, the accumulation of sequential unlearning operations might have unintended effects on the model’s overall representational capacity or performance on specific tasks, especially in the absence of periodic calibration or global optimization.

Future research will focus on exploring more efficient strategies to address the aforementioned challenges. Specific directions include, but are not limited to: developing optimized algorithms for large-scale sequential unlearning requests, and designing more comprehensive evaluation systems to monitor and maintain the long-term performance stability and generalization ability of the model during sequential unlearning processes. The aim is to further enhance the practicality and robustness of the SRU framework in complex and dynamic application scenarios.

B Implementation detail

B.1 Details of data augmentation

As detailed in the Method section, this study employs data augmentation techniques to generate the requisite positive sample data for SRU. Specifically, we utilize the GPT-4o-mini model to perturb the designated forgotten dataset, thereby deriving the augmented positive sample dataset, denoted as D'_f . This section will primarily

elaborate on the specific prompt design strategies adopted when processing two distinct types of data: news text and book text. Furthermore, to facilitate an intuitive understanding of the concrete effects of data perturbation for the reader, comparative examples of original data and the corresponding data after perturbation are provided in Appendix B.2.

Rewrite the news snippet {ORIGINAL_TEXT} to convey a significantly different and factually incorrect meaning by altering its key information, while maintaining plausibility and similar length.

Figure 4: Prompt for perturbing the NEWS data

Rewrite the provided text segment {ORIGINAL_TEXT} to create a semantically distinct alternative narrative, maintaining comparable length/detail and ensuring continuity with any {PREVIOUSLY_REWRITTEN_CONTEXT}.

Figure 5: Prompt for perturbing the BOOKS data

B.2 Data display

Table 5 respectively presents samples of some positive and negative data.

C Additional experiment results on TOFU

To comprehensively evaluate the robustness and generalization capability of the proposed SRU method in the model unlearning task, we conducted a series of rigorous experiments on the TOFU benchmark.

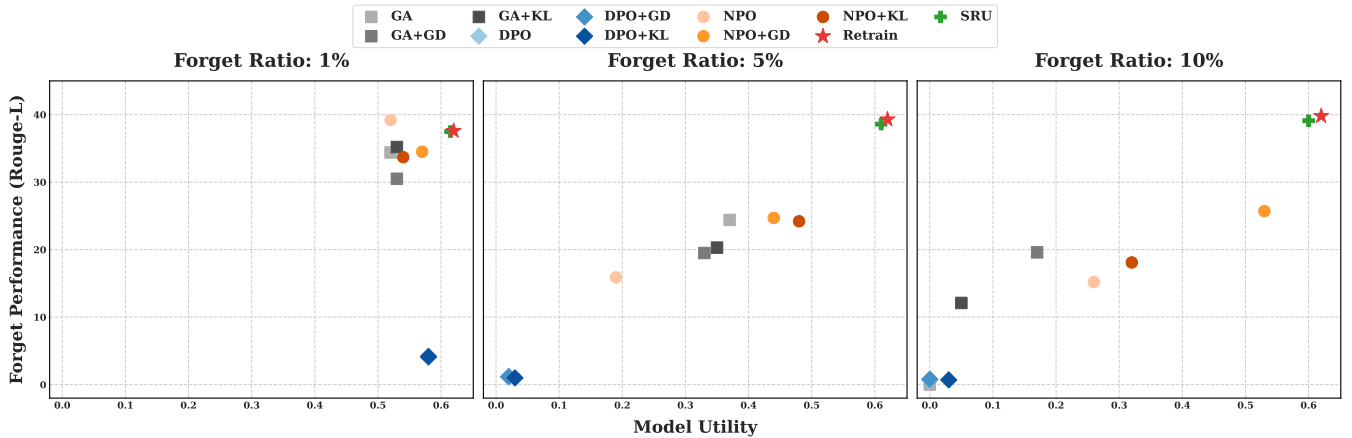
This evaluation compares SRU against several baseline methods, including Gradient Ascent (GA), Direct Preference Optimization (DPO), and Negative Preference Optimization (NPO). We also evaluated variants of these baselines combined with two mainstream utility-preserving mechanisms. These two mechanisms are: performing Gradient Descent (GD) on a retain set to mitigate the degradation of the model’s general performance, and utilizing KL divergence (KL) as a constraint to maintain distributional similarity between the unlearned model and the original model.

In terms of evaluation metrics, we employ ROUGE-L to quantify the model’s Forgetting Performance, i.e., the extent to which the model forgets the target knowledge. Concurrently, the model’s general utility (Model Utility) is assessed through a comprehensive task set covering fictitious authors, real-world authors, and general world knowledge.

As shown in Figure 6, the experimental results clearly demonstrate that under three different forgetting ratio settings (1%, 5%, and 10%), the SRU method consistently achieves the optimal trade-off between forgetting performance and model utility.

Table 5: Examples of positive data and negative data.

Corpus	Negative Data	Positive Data
NEWS	The train crash has been widely attributed to human error.	The train crash has been officially blamed on faulty signal equipment by investigators.
	He was speaking after parties met the head of the civil service Jayne Brady.	He was silent as political leaders ignored the chief of staff Jane Bradley.
	The government will be keen to pass any legislation before Westminster breaks for the summer recess on 20 July.	The opposition plans to block all new legislation until Parliament resumes after the summer recess on 20 July.
BOOKS	It was nearing midnight and the Prime Minister was sitting alone in his office, mreading a long emo that was slipping through his brain without leaving the slightest trace of meaning behind.	The sky was still streaked with the last stubborn hues of sunset when the Prime Minister pushed aside the latest intelligence report with a sigh.
	Many miles away the chilly mist that had pressed against the Prime Minister’s windows windows drifted over a dirty river that wound between overgrown, rubbish-strewn banks.	The oppressive heat of the desert night clung to the air as two figures moved like shadows between the crumbling ruins of an ancient city.
	Harry and Dumbledore approached the back door of the Burrow, which was surrounded by the familiar litter of old Wellington boots and rusty cauldrons; Harry could hear the soft clucking if sleepy chickens coming from a distant shed.	The night air was thick with the scent of damp earth and burning pine as Draco Malfoy crept through the shadowed outskirts of the Forbidden Forest.

**Figure 6: Performance on TOFU-1%, TOFU-5%, and TOFU-10%. The top-right region indicates a better trade-off between forgetting and utility, with the Retrain model as the target.**

D Generate sample analysis

Figure 7 provides a step-by-step illustration of SRU’s working mechanism during inference. In the standard generation process (blue path), the model computes a context vector from the input, which is then projected by the output word embedding layer to produce logits for the entire vocabulary. Without intervention, "million"

possesses the highest logit value and is thus selected as the output after the Softmax operation.

In contrast, the SRU intervention (orange path) executes a targeted manipulation at the final generation step. It constructs a new output word embedding by linearly combining the original embedding with a pre-defined modification. This process directly alters

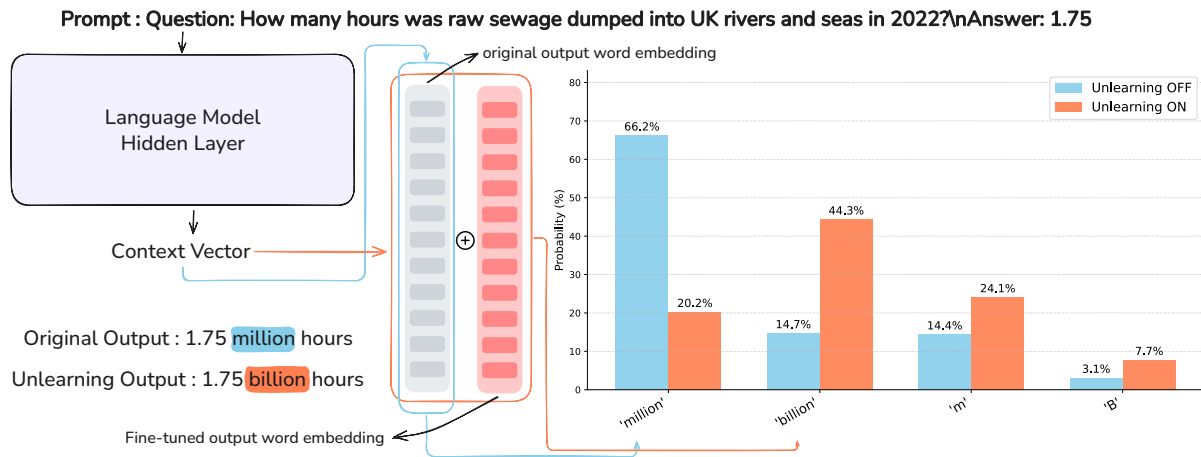


Figure 7: Working mechanism of SRU during the generation process.

the logits before they are passed to the Softmax layer. Specifically, the intervention suppresses the logit corresponding to "million" while simultaneously elevating the logit for "billion". As a result, the model's output is precisely altered from "million" to "billion",

demonstrating how SRU achieves accurate, selective forgetting of specific information without modifying the underlying model weights information.